Candidates' Name

……..…….....

Signature: ……..

Random No.					Personal No.		

(Do not write your school / Center name or Number anywhere)

physics

P510/2

Theory

July / Aug, 2026

3 hours

KAMTEC EXAMINATIONS BOARD

Uganda Advanced Certificate of Education

PHYSICS

PRACTICAL

Paper 2

3HOURS

INSTRUCTIONS TO CANDIDATES:

-This examination paper consists of **two compulsory** items.

-Use **blue** or **black** ink. Any work done in pencil except graphs and diagrams will not be scored.

- Scores are awarded mainly for a clear record of the observations actually made, for their suitability, accuracy and for the use made of them.

-Details on the question paper should not be repeated in the responses.

-Graph paper is provided.

-Silent non-programmable scientific calculators may be used.

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Item 1

1. A game ranger regularly uses a telescope to monitor grazing animals and possible intruders across the open grasslands and nearby hills. During one of the patrols, the telescope accidentally fell onto a rock and eventually stopped working making it difficult to continue with the surveillance activities. On inspection, it was discovered that the concave lens that serves as the eyepiece was damaged. A concave lens with a focal length in the range -14.0 cm to - 17.0 cm is required to repair the telescope. One of the staff found an unlabelled concave lens in the store and suspected it to be suitable for fixing the telescope. The ranger was not convinced and therefore wished to establish the suitability of the lens obtained before installation. The ranger was advised by the senior staff to verify the necessary optical property of the lens using two methods for accuracy.

Method I: Using a convex lens, plane mirror, an illuminated object, and a metre rule, taking a single measurement.

Method II: Using an illuminated object, metre rule and a lens combination, taking several measurements then the property of the lens determined graphically. The ranger did not know how to go about the experimental procedures and has approached you for help.

Task

Plan and carry out an investigation using the two methods to help the ranger decide whether the lens is appropriate for fixing the telescope.

Item 2

A school planned to acquire rolls of constantan wire for use in physics practical examinations. Candidates will be required to use pieces of resistance wires of resistivity between 6.5 x 10^{-7} Ωm and 9.0 x 10^{-7} Ωm. A laboratory technician obtained a roll of bare wire from a neighbouring school but the wire had no label, racing concern for its suitability. The technician eventually designed an investigation using two methods. The results from the two methods were required for obtaining a value which was more accurate. However, due to busy schedule the technician was not able to carry out the investigation and has approached you with the piece of the constantan wire for help.

Task

Follow the procedure below basing on the scenario and use the result to provide feedback to the laboratory technician.

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Method I

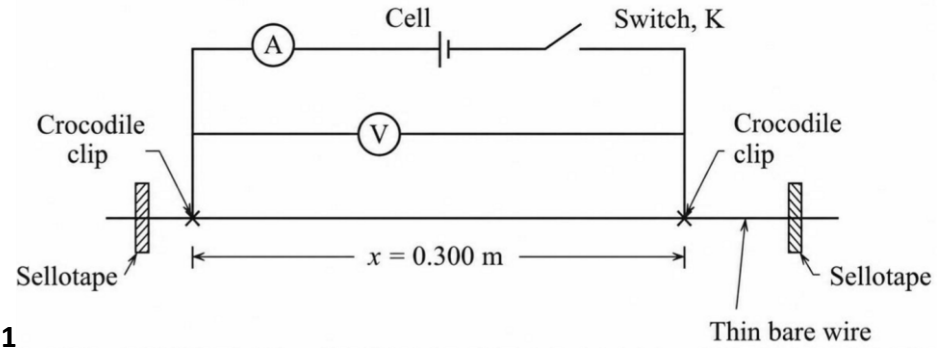


Fig.1

(a) measure and record the diameter,D of the bare wire provided, in metres.

(b) Connect the circuit as shown in the figure 1.

(c) Adjust the length $x\ to\ 0.300\ m$, Close switch, K.

(d) Read and record the ammeter and voltmeter readings, $I\ and\ V$, respectively.

(e) Open the switch.

(f) Calculate the resistivity $\rho_1\ of\ the\ wire\ from\ the\ expression$; $xI\rho_1 = 0.25V\pi D^2$

Method II

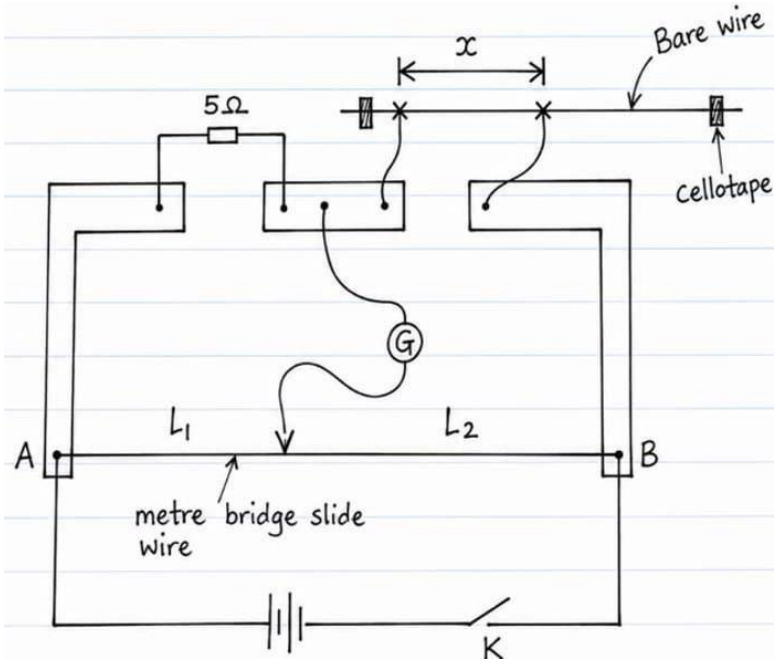


Fig.2.

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(a) Connect the circuit as shown in the figure 2.

(b) Adjust the length of the bare wire provided to $x = 0.200m$.

(c) Close the switch and locate the balance point, C, along the bridge wire AB.

(d) Read and record (in metres), the balance lengths, L_1 and L_2.

(e) Open the switch.

(f) Repeat steps (b) to (e) for other values of $x = 0.300, 0.400, 0.500, 0.600 \; and \; 0.700 \; m$.

(g) Tabulate the results including values of $\frac{L_1}{L_2}$ and $\frac{5}{x}$

(h) State sources of errors and their corresponding precautions taken to mitigate them.

(i) Plot a graph of $\frac{5}{x}$ against $\frac{L_1}{L_2}$

(j) Determine the slope S of the graph.

(k) Determine resistivity ρ_2 of the wire using $S \; and \; D$ obtained.

(l) Obtain the more accurate value of the resistivity ρ of the bare wire provided and give feedback to the school on the suitability of the rolls bare wire supplied.

THE END

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